



SYSTEM FLOW – Driving Lesson Booking System

1. Instructor Setup Phase

1.1 Create Availability

The instructor creates available time slots for lessons.

Each availability belongs to one instructor

One instructor can create many availability slots

Foreign Key:

Plain text

instructor_id

1.2 Create Lesson Offerings

The instructor creates lesson types (e.g., beginner, advanced).

Each lesson offering belongs to one instructor

One instructor can create many lesson offerings

Foreign Key:

Plain text

instructor_id

2. Booking Phase (Main Process)

2.1 Student Makes a Booking

The student selects:

an instructor

a time slot (availability)

a lesson offering

A booking is created to store all this information.

Foreign Keys in Bookings:

Plain text

student_id

instructor_id

availability_id

offer_id

2.2 Meaning of Booking Relationships

One student can make many bookings

One instructor can receive many bookings

Each booking belongs to exactly one student and one instructor

3. Availability Usage

Each availability slot can be used only once

Some slots may remain unused

4. Payment Phase

After booking, the student uploads proof of payment.

Each payment belongs to one booking

A booking may or may not have a payment

Foreign Key:

Plain text

booking_id

5. Lesson Progress Phase

During lessons, the instructor records progress and feedback.

One booking can have multiple progress records

This supports:

viewing lesson progress

instructor feedback

Foreign Key:

Plain text

booking_id

6. Review Phase

After completing the lesson, the student can leave a review.

Each review belongs to one booking

A booking can have at most one review

Foreign Key:

Plain text

booking_id

 **KEY IDEA**

The Booking entity is the center of the system.

Everything connects through it:

Payment

Lesson Progress

Review

SUMMARY OF ALL FOREIGN KEYS

Plain text

Availability:

instructor_id

LessonOffering:

instructor_id

Bookings:

student_id

instructor_id

availability_id

offer_id

Payment:

booking_id

Review:

booking_id

LessonProgress:

booking_id

SIMPLE EXPLANATION

The system works around bookings. A student selects an instructor, a lesson, and a time slot. That creates a booking. After that, payment, lesson progress, and reviews are all linked to that booking.

The instructor can view and respond to booking requests using the Bookings entity.

When a student creates a booking, it is stored with a status such as "Pending". The instructor can view all bookings assigned to them by using the instructor_id.

To respond to a booking request, the instructor updates the booking status. For example, the status may change from "Pending" to "Accepted" or "Rejected".

This means that both "View Booking Requests" and "Respond to Booking Request" are handled through the Bookings entity.